BrewCore

Portable Stability

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Multi-Usage in Mobile Living

Project Overview

A modular brewing system designed for caravans and mobile living environments.
BrewCore integrates water heating, tea brewing and coffee preparation into a compact vertical architecture.

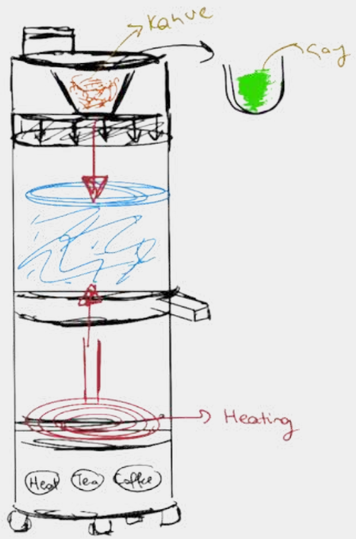
Context
Preparing hot beverages in mobile environments such as caravans areas presents several challenges:

- limited kitchen space and surfaces
- the need for multiple appliances

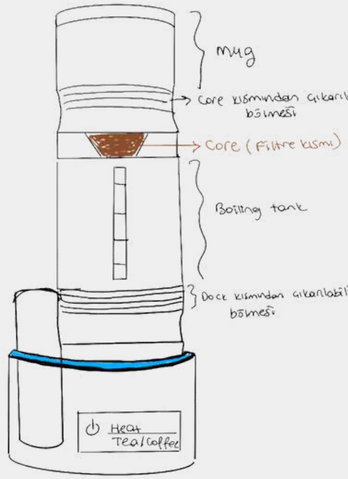
Design Goal
The goal of the project is to design a portable brewing system that combines these functions within a modular product that can be easily assembled, disassembled, and customized according to user needs.

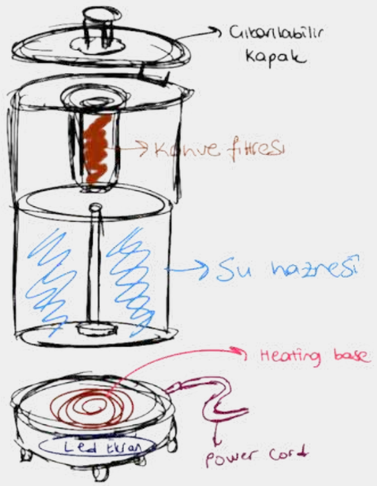
Keywords
Portable
Multi-usage
Mobile living
Compact footprint

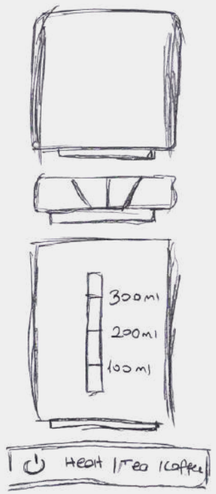
Key Design Milestones (Process Timeline)



Problem Definition
Understanding the challenges of preparing hot beverages in caravan environments.
↓
Research Phase
Investigating user needs related to portability, multifunctionality, and stability.
↓
Direction Decision
Focusing the project on a modular brewing system designed specifically for mobile environments.
↓
Concept Exploration
Developing several structural alternatives to explore heating and brewing integration.
↓
Concept Selection
Selecting a vertical modular architecture due to its compact footprint and adaptability.
↓
Prototype Development
Testing the stacking logic and brewing flow through a quick prototype.
↓
System Development
Refining the modular architecture and heating system of the product.







Research Insights / Concept Exploration

Limited Space
Mobile kitchens require compact appliances with minimal footprint.
Multi-Usage Expectation
Users prefer one device capable of preparing tea, coffee, and hot water instead of carrying multiple appliances.
User Control
Users feel safer when they can visually monitor the water level and heating process.
These insights defined the main design drivers of the project.

Selection Criteria
Smallest footprint
Clear modular architecture
Compatible with mobile environments
Simple brewing flow
Rejected Alternatives
Concept B
Less stable structure.
Concept C
Larger footprint and less portable.
System Architecture
Final system includes four components:
Dock — heating base with control interface
Boiling Tank — water heating chamber
Core — interchangeable tea or coffee filter module
Mug — detachable drinking cup

Tea, Coffee & Hot Water
in One Device

Design Decisions

What I Accepted

• A vertical modular architecture to reduce countertop footprint in mobile environments.
• A four part system structure consisting of Dock, Boiling Tank, Core (filter module), and Mug.
• Interchangeable brewing modules allowing both tea and coffee preparation.
• A magnetic alignment dock system that guides the boiling tank into position for heating.
• A double wall insulated boiling tank structure to maintain temperature while keeping the outer surface safe to touch.

What I Rejected

• Large horizontal brewing systems that require more countertop space.
• Separate appliances for tea, coffee and hot water preparation.
• Fixed brewing systems where components cannot be detached.
• Side spout dispensing systems that complicate the modular structure.

What I Transformed

• The traditional electric kettle concept into a modular brewing system.
• A standard heating appliance into a multi-usage brewing device capable of preparing tea, coffee, or hot water.
• A fixed kitchen appliance into a portable system suitable for mobile living environments.
• A single size boiling tank into a customizable system with multiple capacity options.

Design Decisions

Key Design Decisions
Adopt a vertically stacked modular architecture.
Ensure all components can be attached and detached easily.
Create interchangeable brewing modules for tea and coffee preparation.
Allow users to configure the system based on their needs.
Design a detachable drinking cup integrated into the system.
Provide multiple boiling tank capacity options.
Product Customization
The system allows users to choose different boiling tank capacities depending on their usage needs.
Three capacity options are available:
- **small boiling tank capacity**
- **medium boiling tank capacity**
- **large boiling tank capacity**

This allows the system to adapt to different usage scenarios, from individual use to small group preparation.

150 ml 250 ml 350 ml

Engineering & Material Strategy

All components are designed to connect through a modular stacking system, allowing users to assemble or detach modules easily.

Heating System

The dock includes a magnetic alignment heating base.

Magnets guide the boiling tank into the correct position while concealed electrical contacts transfer energy to the internal heating element.

This system allows:

- secure positioning
- easy detachment
- safe power transfer

Material Structure

The boiling tank uses a double-wall insulated structure commonly used in high-performance thermal containers.

Inner Chamber

Food grade stainless steel ensures durability and heat resistance.

Insulation Layer

A vacuum insulated layer reduces heat transfer and maintains internal temperature.

Outer Shell

A heat-resistant polymer or metal housing protects the user from external heat.

This structure ensures that the outer surface remains safe to touch while maintaining high internal temperature.

Stability Strategy

To improve stability in mobile environments, the dock includes:

- silicone anti slip base
- optional suction grip
- low center of gravity

These features help prevent movement or sliding in caravans.

Color Options

Outdoor Neutral

Earth Tone

Deep Carbon

Exploded Views

To heat water

To brew tea or coffee

AI Interaction & Reflection

Selected Impactful Prompts

“How do people prepare hot beverages in camping/caravans environments?”
“What design considerations improve stability in portable appliances?”
“How can modular systems integrate tea and coffee brewing?”

AI Role Tags

Clarifier
Explained user contexts and environmental constraints.
Re-framer
Helped redefine the design direction toward portability and multi-usage.
Expander
Generated alternative concept possibilities.
Critique
Supported evaluation of concept alternatives.

AI Response Summary

AI responses helped identify compact appliance strategies, modular system structures, and heating system possibilities. These insights supported concept exploration and decision making.

Reflection

AI tools supported the design process by helping to clarify ideas and explore alternative directions. However, the final design concept and product architecture were decided upon by the designer through iterative sketching, prototyping, and design evaluation.